

How This System Works

Strategic Intelligence — Banking & Payments

Printec Group · July 2026

Purpose of this document

This explains **how the system reaches its conclusions**, so that you can judge them rather than trust them. It follows the work step by step. At each step it says what happens, which established method we follow, where that method comes from, and what can go wrong.

No background is assumed. Specialist words are explained where they appear, and there is a glossary at the end.

Two things you will not find here. There are no volume figures — how many sources or documents we hold — because this is about method, not size. And there are no accuracy figures, because the system has never been scored against real outcomes. That gap is real and is discussed in step 12.

Three words are used throughout, and they mean different things:

Software	Conventional computer code. Follows fixed rules and gives the same answer every time.
AI	An artificial-intelligence language model — software that reads and writes text, and can make judgements. It can also be wrong in ways ordinary software cannot.
Researcher	An AI working autonomously on a task, able to search and read on its own.

The rule that shapes the whole design: **AI proposes, software decides**. Wherever a judgement can be checked mechanically, ordinary software does the checking. The AI is never trusted to confirm its own work.

Part A — Where our methods come from

Why a banking research system borrows from intelligence work

Market research and intelligence analysis answer the same kind of question: *what is happening out there, what does it mean, and what is likely next* — from incomplete public information, where being confidently wrong is expensive.

The difference is that intelligence services have spent decades writing down how to do that reasoning honestly, and their methods are public.

The reason they wrote them down matters to us. In 2003, US and UK intelligence assessed that Iraq held active weapons programmes. That assessment was wrong, and the subsequent inquiries found the cause was not bad information-gathering but **bad reasoning**: assumptions nobody had written down or questioned, single uncorroborated sources treated as confirmed, and evidence read as supporting the expected answer. In response, the United States passed the **Intelligence Reform and Terrorism Prevention Act of 2004**, which required its agencies to use documented, inspectable analytic methods rather than expert instinct alone. Those

methods were then published.

Those are exactly our failure modes. We also work from fragments, we also risk reading evidence as confirming what we expect, and we also risk treating one story repeated three times as three sources.

Commercial market research has no comparable body of written method. It has plenty on how to *present* conclusions, and very little on how to *reach* them defensibly. So we borrowed from the field that does.

The three sources we draw on

1. The Intelligence Cycle

What it is. The standard description of how an intelligence organisation works, as a repeating loop: decide what you need to know → collect material → put it into usable form → analyse it → deliver it to whoever must act → take their feedback into the next round.

Is this official? It is not one author's framework or a consultancy's product. It is the model taught in published government and military manuals in the United States, the United Kingdom and NATO. It is closer to common professional knowledge in that field than to a proprietary method. Where this document says *doctrine*, it means an officially published, mandatory procedure — not dogma.

Why we use it. It forces collection to be driven by declared needs rather than by whatever is easy to find, and it makes each step separately testable — we can repair or replace one step without rebuilding the others.

Where each step of the cycle appears in this document:

Stage of the cycle	Steps in Part B
Decide what you need to know	1
Collect	2, 3, 4
Put into usable form	5, 6, 7, 8, 9
Analyse	10, 11
Deliver	15
Know your gaps, and feed back	12, 13, 14

2. Early warning from weak signals

What it is. A way of spotting change while the evidence is still fragmentary, rather than waiting for confirmation. Its central idea is the **weak signal**: an early, incomplete, ambiguous indication that something is shifting. The argument is that by the time a signal is strong and clear, the time in which you could have acted has gone.

Where it comes from. Corporate strategy, not intelligence. Introduced by **Igor Ansoff**, a management professor widely regarded as the founder of strategic management as an academic field, in an article called *Managing Strategic Surprise by Response to Weak Signals*, published in 1975 in

California Management Review. That journal is **peer-reviewed** — meaning independent experts examined the work before it was published, rather than the author publishing it himself.

Why we use it. This is the product. A published tender is a strong signal: every competitor sees it the same morning. A newly appointed technology chief, a merger that will force two card platforms into one, or a capability adopted by comparable banks in nearby markets — those are weak signals, and that is where an

advantage exists.

3. Structured analytic techniques

What they are. A published catalogue of procedures for reasoning from evidence: listing alternative explanations before choosing one, writing down your assumptions, rating how good your sources are, deciding in advance what would prove you wrong, and having someone deliberately argue the opposite case. Their purpose is **not** to make analysts cleverer. It is to force reasoning into the open where colleagues can inspect and challenge it.

Where they come from — all publicly available:

- **A Tradecraft Primer** — published by the **CIA's Center for the Study of Intelligence** in March 2009. Public domain, meaning free of copyright and freely reproducible. Describes twelve core techniques. "*Tradecraft*" simply means the professional craft of the job.
- **Structured Analytic Techniques for Intelligence Analysis** by **Richards Heuer** and **Randolph Pherson**, published by SAGE, third edition 2020 — the standard textbook, cataloguing sixty-six techniques. Heuer spent his career at the CIA; his earlier book on the psychology of analysis is the foundational text of the field.
- **Intelligence Community Directive 203** — titled *Analytic Standards*, an official directive of the US Office of the Director of National Intelligence, setting rules that all US intelligence analysis must follow, including how sources must be described and how uncertainty must be expressed. Referred to below as **ICD 203**.

An honest caveat — and why we still use them

These methods are **not proven to make conclusions more accurate**.

A controlled experiment with fifty professional intelligence analysts tested the best-known technique and found no improvement in accuracy at all — and that analysts became *less consistent*, meaning the same analyst shown the same evidence twice reached different conclusions. A review by the **RAND Corporation**, an independent American research institute, found that US agencies require these methods while making little effort to measure whether they work.

So why use them? Because accuracy is not the only thing we need, and it is not the thing we can get first. What these methods deliver is that a conclusion can be **traced, questioned and overturned** — you can see which evidence produced it, which assumptions it rests on, and what would disprove it. A confident conclusion nobody can interrogate is worth less to a bank than a hedged one they can take apart.

They also make accuracy *measurable later*: a system that records its reasoning can be scored against what actually happened. One that does not, cannot. We are not there yet — see step 12.

Part B — What happens at each step

1. Deciding what to look for

What happens. Broad questions are broken into specific things worth watching, and each of those into a concrete instruction: find sources of this kind, in this country, in this language.

Method — established. In intelligence practice this is called **collection requirements management**. A broad question becomes a list of observable things that would answer it, and each of those becomes a specific task someone can be assigned. Collection is never opportunistic. *Source: standard published government and military manuals.*

Why. It is the difference between a professional researcher and an automated scraper — a program that simply downloads whatever it finds. It also makes gaps measurable: you can only report a gap against a need

you actually wrote down.

In our case. Six standing themes, each with declared topics, languages and refresh frequency, per country: company disclosures · tenders and procurement · regulation and deadlines · market statistics · early signs of intent such as hiring · vendor and competitor activity.

Example. Under "regulation and deadlines" for Greece, one declared need is: *rules that oblige payment providers to do something by a stated date*. That produces the instruction to monitor the Bank of Greece and the European Central Bank, in Greek and English.

2. Finding sources worth monitoring

What happens. For each declared need, an AI researcher works out where a professional analyst covering that market would actually look — in the local language first — and proposes specific sources, justifying each one.

Method — ours. Published practice assumes either a human analyst who already knows the territory, or a search engine ranking by popularity. Neither works here. The authoritative sources in our markets are national procurement portals, central bank publications and local-language trade press. Popularity rankings do not surface them, and a general search will not find them without already knowing they exist.

How it works. The researcher reasons **from the market itself**, not from keywords: who regulates this market, who is legally obliged to publish, who buys, who sells, who reports on them, and in which language. Each proposed source must state which declared need it answers. Nothing is monitored merely because it appeared in a search result.

Example. For tenders in Greece: public bodies must publish contract awards → Greece has a central procurement portal and a government transparency portal → but banks are not public bodies, so their tenders appear only on their own supplier pages → therefore monitor both, in Greek.

Why we built our own. There is no published method for automated, multilingual, needs-driven source discovery in a commercial market.

3. Deciding what to trust

What happens. A proposed source is assessed before it is monitored: what it is, who publishes it, what it covers, in what language, how often it changes, and whether we may lawfully collect from it.

Method — established. Intelligence practice grades the reliability of the **source** and the credibility of the **information** as two separate judgements — a normally reliable outlet can still report something doubtful. *Source: standard practice across NATO and allied services.*

Why. Later, our confidence in a conclusion depends on how many *independent* sources support it. If we cannot say what a source is, we cannot say whether two sources agreeing means anything.

Current state — honest. We record what a source is, but we do **not** yet rank sources by standing. A regulator's official filing should outrank a company's own announcement, which should outrank a trade publication, which should outrank a website that merely republishes that publication. Today all four count equally, which flatters widely-copied stories. This is specified and not built.

4. Collecting documents

What happens. Documents are fetched, stored, and checked to see whether they have genuinely changed since last time — so a page republished unchanged is not recorded as news. Documents that are scans rather than text are read by character-recognition software.

Method — ordinary software engineering, not an analytic method, and we do not claim otherwise.

Why it matters for the conclusions. Without change detection, a website refreshing a page looks like market activity. The system would report movement where there is none.

One deliberate rule. Some websites assemble their pages using scripts, and must be read by an automated browser rather than a simple download. If that browser is unavailable, **collection stops** rather than storing empty pages. An empty document that looks real is worse than a missing one, because it silently becomes evidence that nothing is happening.

5. Turning documents into facts

What happens. An AI reads each document and proposes facts in a simple form — *organisation* → *did what* → *to whom* — each with the exact sentence that supports it. Ordinary software then checks each proposed fact against the original document before accepting it.

Method — established principle, our enforcement. The principle is **attribution to identified sources**, required by ICD 203: every assertion must be attributed to a source, described well enough that a reader can judge it independently.

What we added is enforcement. The standard asks analysts to attribute properly. Our system rejects a proposed fact unless its supporting sentence appears **word for word** in the source document. That check is done by ordinary software, not by the AI — the AI is never asked to confirm its own honesty. Spacing and accent marks are ignored so formatting cannot cause false rejections; the words must match. The check works in the source's own language: a Greek document keeps its Greek sentence, and Greek is compared with Greek. Nothing is translated before checking.

Example. An AI proposing "*Bank X is planning to replace its core banking system*" from a document that says only "*Bank X reported strong digital growth*" is rejected immediately — that sentence does not exist in the document. The fact never enters the system.

What this does and does not guarantee — important. It proves the quoted sentence is **real**. It does **not** prove the sentence actually supports the conclusion drawn from it. An AI can quote a genuine sentence and attach an inference the sentence does not justify. This check eliminates invented quotations; it does not eliminate faulty reasoning from real ones. That second problem is the job of step 14, and step 14 is the weaker of the two.

6. Recognising the same organisation twice

What happens. Every reference to one organisation is combined into a single entry — across languages, legal suffixes and abbreviations — so *Alpha Bank*, *Alpha Bank S.A.* and the Greek spelling are one company, not three.

Method — established. Known as **record linkage**: the statistical discipline of deciding when two records refer to the same real-world thing. *Source: a 1969 paper by Ivan Fellegi and Alan Sunter in the Journal of the American Statistical Association. The same method underpins census, health-service and customer-data matching worldwide.*

How we do it. By fixed rules: match on an official identifier — a tax number or legal entity identifier — where one exists, and only fall back to a standardised name, within one country, where none does.

Why not match by meaning. We tested matching organisations by how similar their names *mean*, using the same AI technique that powers search-by-meaning. It failed badly: it proposed merging

Visa with Mastercard, because two short company names in the same industry are very close in meaning even though they are rivals. Official identifiers cannot fail that way.

Honest limitation. Official identifiers are rarely quoted in the sources we read, so in practice almost all matching is by name — and we have not measured how accurate that is across Greek and English.

7. Establishing when and where something happened

What happens. We record the date and country of **the event itself**, kept separate from when and where we collected the document.

Method — established. Called **bi-temporal recording**, which simply means keeping two independent clocks: when something was true in the world, and when we came to know it. *Source: a standard database design discipline set out in Richard Snodgrass's Developing Time-Oriented Database Applications (1999). Intelligence practice makes the same distinction between when an event occurred and when it was reported.*

Why. With one clock, an event from two years ago that we found last week reads as news — and the question “*what has changed since I last looked?*” cannot be answered at all.

A rule learned the hard way. A fact's date and country describe the **event**, never the collection. A story found on a Greek website about a bank in Bulgaria is a Bulgarian fact. Where the location cannot be established, the fact says *unknown* rather than quietly defaulting to the market being collected.

Example. “*Mashreq launched a digital account for visitors*” — Mashreq is a bank in the United Arab Emirates, and the story was found in a Greek trade publication. It is recorded as a 2026 UAE event. Filing it as Greek would have been both wrong and useless.

8. Separating events from background

What happens. Facts describing something *happening* — an acquisition, a contract award, a product launch — are marked as distinct from standing description, such as which regulator supervises which bank.

Method — established. In intelligence practice an **indicator** is an observable event decided in advance to be meaningful; an **indication** is an actual occurrence of one. *Source: the Tradecraft Primer, where it appears as “Indicators or Signposts of Change”.*

Why. Things that happen carry forward-looking meaning. A directory entry does not.

A correction worth recording. We originally let only these events reach the analysis steps. That was a mistake: it hid regional expansion, regulatory obligations and corporate announcements from the very steps whose job is to notice indirect signals. All analysis steps now read **everything**. The marking is a label to help a reader filter, never a filter on what the system may consider.

9. Counting corroboration honestly

What happens. Several reports of the same event are collapsed into one fact recording how many

independent documents support it — while genuinely different events stay separate.

Method — established. Guarding against **circular reporting**: the recognised failure in which one original report reappears in many places and is mistaken for several independent confirmations.

Source: standard collection doctrine.

Why. Three trade articles rewriting one press release are **one** source, not three. Without this, apparent confidence measures how widely a story was copied rather than how well established it is.

In our case. Two facts are treated as the same event when the organisation, the relationship, the other party and the **month** all match. The month is deliberately part of that test, so the same two companies doing the same thing two years later stays a separate event.

Current state — honest. We do not yet compare document *text* for near-identical wording. The same news-wire story appearing on two websites still counts as two independent documents. This is the most common form of the failure and it is not yet closed.

10. Working out what an organisation will need

What happens. Everything one organisation has done is read together, and the system states what it will consequently require — as a business need, independent of any supplier who might meet it.

Method — established. The indicators method above, applied forward. Each inference takes the form

trigger → consequence → time window. A completed merger implies consolidating two platforms into one, typically within one to three years. A regulation binding a class of institution implies an obligation on every member of that class, by the stated deadline. A contract reaching the end of its term implies a re-tender, typically three to twelve months beforehand.

Why this rather than direct judgement. It makes the reasoning inspectable. A reader sees which trigger fired, and can argue with the trigger rather than with an opaque conclusion.

Each trigger is rated for whether it genuinely discriminates. This is the key discipline. A completed merger **strongly** implies consolidation — consolidation nearly always follows. A new executive appointment only **weakly** implies a change programme — executives are appointed constantly and most appointments precede nothing. Weak triggers may support a conclusion alongside other evidence; they may never carry one alone.

Example of the rule working. A need inferred purely from someone's promotion to Head of Commercial Banking Strategy was rejected by the system: nothing else in the evidence indicated a gap or a programme. The pattern is real; on its own it proves nothing.

Specified, not built — the indicator test. A formal check of each trigger against the question

"would we see this even if the conclusion were false?" A trigger that would appear regardless is worthless however sensible it sounds. *Source: Randolph Pherson, 2008.* We apply this judgement by hand rather than as a defined step.

11. Anticipating what nobody has announced

What happens. Things already done successfully in other countries are described as repeatable patterns, and comparable organisations here are checked for any sign of having done them.

Method — established, two of them.

Structured analogies means: rather than speculating about what an organisation will do, find comparable cases elsewhere, describe them, and derive the expectation from what those organisations actually did. *Source: Kesten Green and J. Scott Armstrong, Structured Analogies for Forecasting,*

International Journal of Forecasting, 2007 — a peer-reviewed forecasting journal. Their research found that structured use of analogies substantially outperformed expert judgement alone.

Diffusion of innovations is the study of how new practices spread through a population in observable stages. *Source: Everett Rogers, 1962 — among the most cited works in social science.*

Why this is the valuable half. A published tender is visible to every competitor the same morning. A capability adopted by comparable institutions in four other countries and absent here is visible only to someone holding a picture that spans both.

In our case. A pattern is stated **without the organisation that made it** — *"a retail bank that had been running its payments business as a joint venture bought out its processor partner"*, rather than naming the bank. How well established a pattern is gets **counted from the evidence, never asserted**: how many separate institutions, in how many countries, actually made the move. A move made once by one institution is an experiment, not a pattern, and cannot produce a conclusion. **Two institutions is the minimum.**

Not our invention, and we should not claim it is. This is a published method with published evidence behind it. Ours is the industrialisation — running the comparison automatically across many countries. That is engineering, not method.

12. Saying clearly what we do not know

What happens. The system refuses to claim an organisation *has not* done something unless it has genuinely examined that organisation. Every refusal is recorded and becomes an instruction to collect more.

Method — ours, built on an established principle. The principle is that a system may only conclude something is absent within the area it has actually looked at; outside that area the honest answer is "we do not know", not "no". Intelligence practice states the principle and keeps a register of collection gaps — but no published method makes the refusal **automatic and measured by evidence**. Ours does.

How it works. Before making any "no sign of this" statement, the system counts how much it holds on that organisation. **Below five separate source documents it does not speak**: the result is recorded as *insufficient coverage*, kept permanently, and the organisation joins a collection priority list. Where it does speak, the wording is exactly what was measured — *"no evidence of this across the N documents we hold on them"* — and a checking step deletes any conclusion phrased as

"they have not done this."

Why this matters more than any other safeguard. Our most valuable conclusion is an absence — this capability is proven elsewhere and this institution shows no sign of it. It is also the most dangerous thing we can say. Tell a customer they lack something they already have, and the system's credibility is gone permanently, taking every other conclusion with it.

Honest limitation. The count includes only documents that already produced a fact about that organisation — so it measures what we successfully extracted, not how hard we looked. A document we read that mentioned the organisation and yielded nothing is the strongest evidence of a genuine absence, and it is invisible to the count. The control is the right shape; the measurement behind it is an approximation.

13. Never repeating work that failed

What happens. The system records that a specific set of evidence was examined — including when it produced nothing, and why. Each record is filed under a digital signature computed from the exact evidence considered, so the same material is recognised if it comes round again.

Method — ours, extending established practice. Intelligence organisations keep a register of unanswered questions to drive future collection. Ours records **analytic attempts**, including those that legitimately produced nothing.

Why. Two completely different situations look identical in a database: *we examined this and found nothing*, and *we never examined this*. Confusing them makes every "nothing found" untrustworthy, and causes the same work to repeat every month forever. We know this because it happened — one part of the system re-examined the same material every single cycle until this record was added.

14. Arguing against our own conclusions

What happens. A second, separate AI attacks each proposed conclusion, arguing against it.

Method — established. Called **devil's advocacy**: a designated challenger argues against the emerging conclusion rather than assessing it neutrally. *Source: the Tradecraft Primer lists it among the "contrarian" techniques; Heuer revisited it in 2008.*

Our addition — the division of labour. Heuer's own criticism is that a lone designated challenger is usually a weak one. Our rule is that **the AI proposes and the software decides**: the challenger may only *keep or delete*, never rewrite a conclusion and never alter the evidence behind it. Its verdict is binding — a rejected conclusion is never published — and the rejection is stored with its reason, so any challenge can be reviewed afterwards. It is told to delete when uncertain, and told why: one over-general pattern would generate a false conclusion against every institution in the market.

Honest limitation, and the most important weakness in the system. The challenger works well in one part and poorly in another. Where it examines the cross-country patterns it rejects roughly **one in three**. Where it examines derived needs it rejects roughly **one in thirty** — and a rejection rate that low is a rubber stamp rather than a control. We have also never sampled what it *accepted*, so we cannot say what it is missing. This step is the main defence against faulty reasoning from real quotations (see step 5), which makes its weakness consequential.

Specified, not built — competing explanations. The best-known technique in the field: list *all* the reasonable explanations, test every piece of evidence against each, and try to **eliminate** explanations rather than confirm a favourite. The most likely explanation is the one with the least evidence against it. *Source: Richards Heuer, CIA.* Today we produce one explanation and check whether the evidence supports it — precisely the mistake this technique exists to prevent. As noted in Part A, we would adopt it to show our working, not on a promise of accuracy.

15. Publishing

What happens. Every conclusion is checked against the underlying evidence before anything is produced for a reader.

Method — established. Intelligence products are reviewed against declared standards before release: is the sourcing described, are judgements distinguished from facts, is uncertainty expressed properly. *Source: ICD 203.*

Our addition — making it stop the process. In human practice this review is advisory: a reviewer comments, the author decides. Ours is automatic and blocking. A conclusion citing evidence that does not exist fails the check, the report is **not** regenerated, and the previous version stands rather than a wrong one being published.

Specified, not built — separating "how likely" from "how sure". ICD 203 requires these to be stated separately and never merged, because they are different: a judgement can be highly likely and rest on thin evidence. We currently publish a single number labelled *confidence*, and it is **not a probability**. It stops rising once two documents support a conclusion, and it never falls to zero. It should be renamed to what it is — a measure of evidence strength — with likelihood stated separately in consistent words.

Specified, not built — scoring ourselves. Conclusions should carry a date by which they can be judged and a description of what would count as being right, then be scored against what actually happened. **Until that exists, nothing distinguishes a system that is right from one that is merely consistent.** This is the single largest gap in the system and it is not a small piece of work.

Part C — Honest summary

What we designed ourselves

Almost everything above is an established method applied as its authors intended. Four things are ours, because nothing published fits:

What	Why nothing published fits
Needs-driven source discovery in local languages (step 2)	Published practice assumes an analyst who knows the territory, or a search ranking — neither surfaces national procurement portals or local-language trade press
Mechanically enforced word-for-word checking (step 5)	The standard asks analysts to attribute properly; nothing makes it impossible not to
Automatic refusal to speak below measured coverage (step 12)	The principle is established; automatic, evidence-measured refusal is not
A record of examinations that found nothing (step 13)	Established practice records unanswered questions, not examinations that legitimately produced no result

Two further places apply an established method with our own twist, and are described as such: the division of labour in the challenge step (14) and making the publication check blocking (15).

And one thing we should not claim. The anticipation machinery in step 11 is not our invention. It is structured analogies, a published and tested forecasting method. It is better that this document says so than that someone recognises the technique and wonders what else has been overstated.

What is specified but not yet built

1. Ranking sources by standing (step 3)
2. Detecting near-identical document text before counting two sources as independent (step 9)
3. The formal indicator test (step 10)
4. Competing explanations rather than one (step 14)
5. Separating likelihood from evidence strength (step 15)
6. Scoring conclusions against what actually happened (step 15)

What the system cannot do

- **It cannot tell you it is right.** Nothing has been scored against outcomes.
- **It cannot tell you about an organisation we have not covered,** and is designed to say so rather than fill the silence.
- **It sees only public information.** No customer records, no internal conversations.
- **It does not know what Printec sells.** Deliberately: a system that knew would keep rediscovering reasons to sell it, and would go blind to indirect openings. A separate step applies Printec's capabilities afterwards, and is currently switched off.

Legal and data-protection position

The system collects only material that is **published and publicly accessible**, respects the access rules websites declare, and limits its request rates. It does not access anything behind a login, and holds no customer data.

It does record **named individuals** where public sources report them — typically executive appointments, which are a legitimate market signal. Under the **General Data Protection Regulation (GDPR)**, the fact that information is public does not by itself permit unlimited processing. Accordingly the system marks records containing personal data, restricts who may see them, and can remove a named individual while keeping the underlying business fact. **Printec's Data Protection Officer owns the policy decisions** — what lawful basis applies, how long records are kept, and what notice is owed. This document describes the mechanism, not the policy.

Glossary

AI — an artificial-intelligence language model; software that reads and writes text and can make judgements, and can be wrong in ways ordinary software cannot.

Bi-temporal — keeping two dates for everything: when it happened, and when we learned it.

Circular reporting — mistaking one original story republished many times for several independent confirmations.

Devil's advocacy — deliberately arguing against your own emerging conclusion.

Doctrine — an officially published, mandatory procedure that a service must follow.

GDPR — the EU General Data Protection Regulation.

ICD 203 — the official US directive setting the standards intelligence analysis must meet.

Indicator — an observable event decided in advance to be meaningful. An **indication** is an actual occurrence of one.

Peer-reviewed — published in a journal where independent experts examined the work first.

Record linkage — deciding when two records refer to the same real-world organisation.

Software — conventional computer code; fixed rules, same answer every time.

Structured analogies — forecasting by finding comparable past cases rather than by judgement.

Tradecraft — the professional craft and methods of a field.

Weak signal — an early, incomplete, ambiguous indication that something is changing.

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